



Breast Cancer Classification

Machine learning workflow to classify tumors as Benign or Malignant

Data Preparation Pipeline

01

Import Libraries

NumPy, Pandas, Matplotlib, Seaborn, Scikit-learn

02

Load Dataset

CSV file with tumor samples and numerical features

03

Clean & Encode

Remove ID column, encode diagnosis (Malignant=1, Benign=0)

04

Select Variables

Features (X) and target variable (y)

Train-Test Split Strategy

80%

Training Data

Used to build and train
classification models

20%

Testing Data

Evaluates model generalization
performance



Feature Scaling

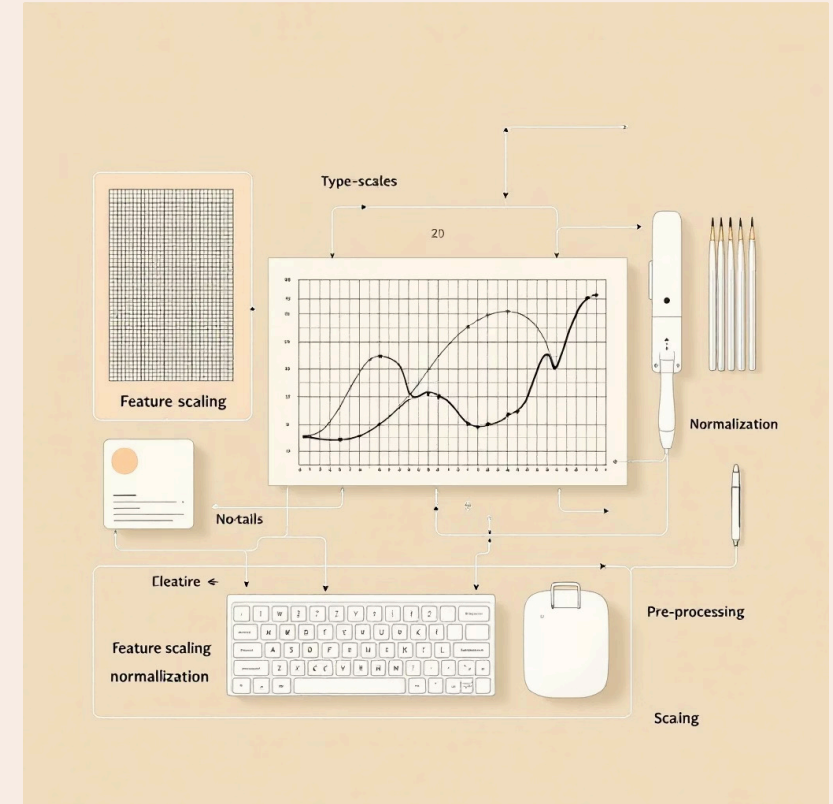
StandardScaler normalizes feature values for optimal model performance

Scaled Models

- Logistic Regression
- Decision Tree
- Random Forest

Unscaled Model

Naive Bayes operates on raw data



Classification Models



Logistic Regression

Linear classifier for binary outcomes



Naive Bayes

Gaussian probabilistic classifier



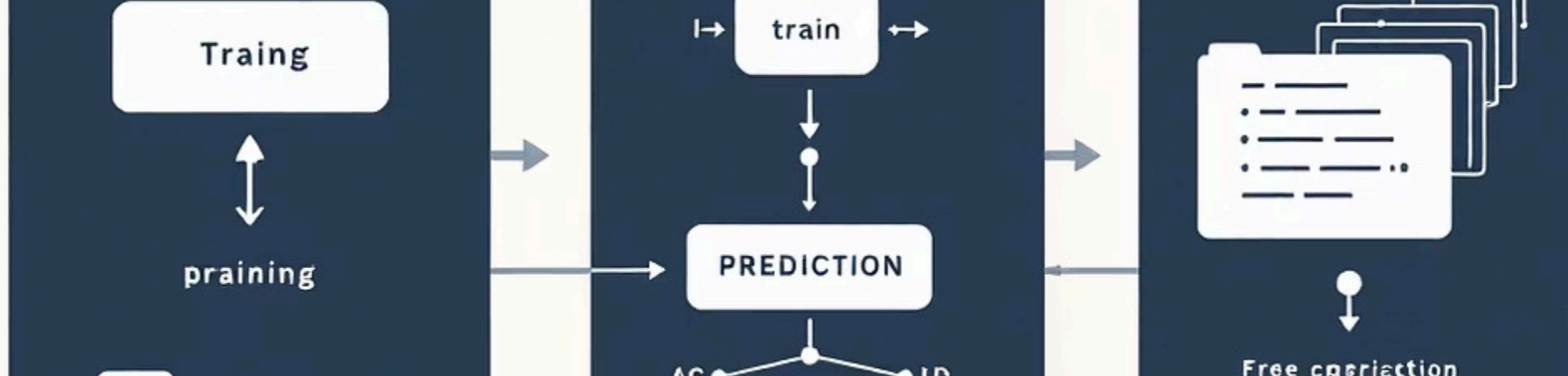
Decision Tree

Hierarchical rule-based classifier



Random Forest

Ensemble of multiple decision trees



Training & Prediction



Model Training

Each model learns patterns from training dataset



Generate Predictions

Models classify test samples and output probabilities

Model Evaluation Metrics

1

Accuracy Score

Overall correctness of predictions

2

Classification Report

Precision, Recall, F1-score breakdown

3

Confusion Matrix

True/false positives and negatives

4

ROC & AUC

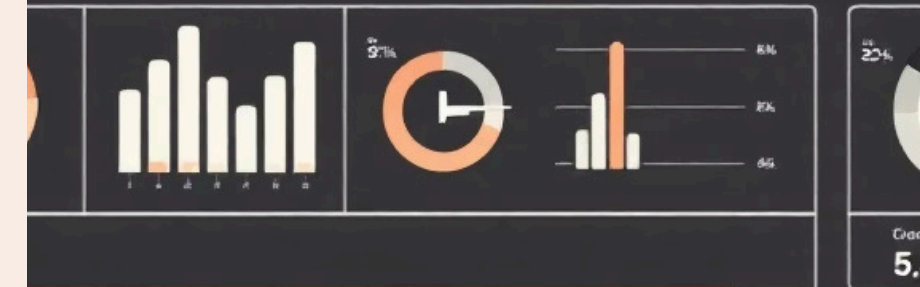
Curve analysis and area under curve score

Evaluation

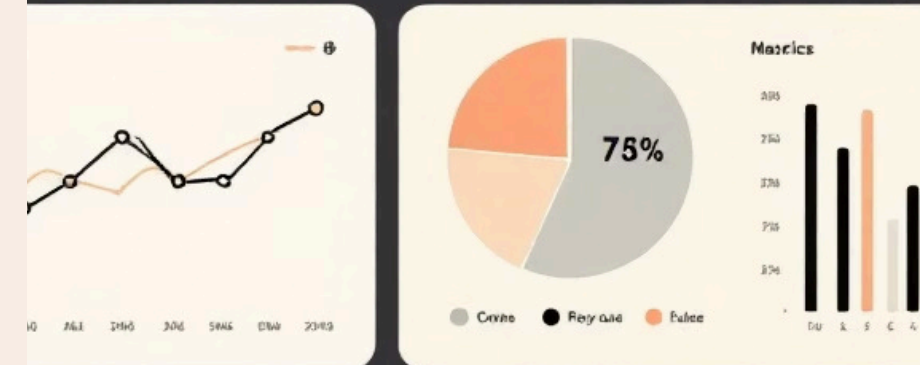
★ ★ ★ Met - 5 20



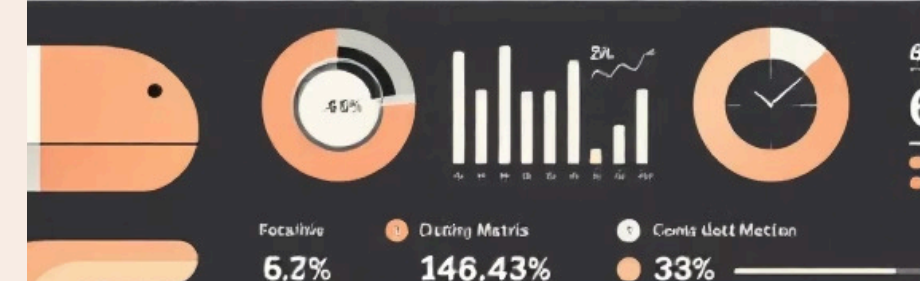
Former
7.0%
① Data, Test Sets



ion in Metrics



Queries



Visualization Insights

- **Accuracy Comparison**

Bar chart showing model performance

- **Confusion Matrices**

Heatmaps for each classifier

- **ROC Curves**

Model comparison visualization

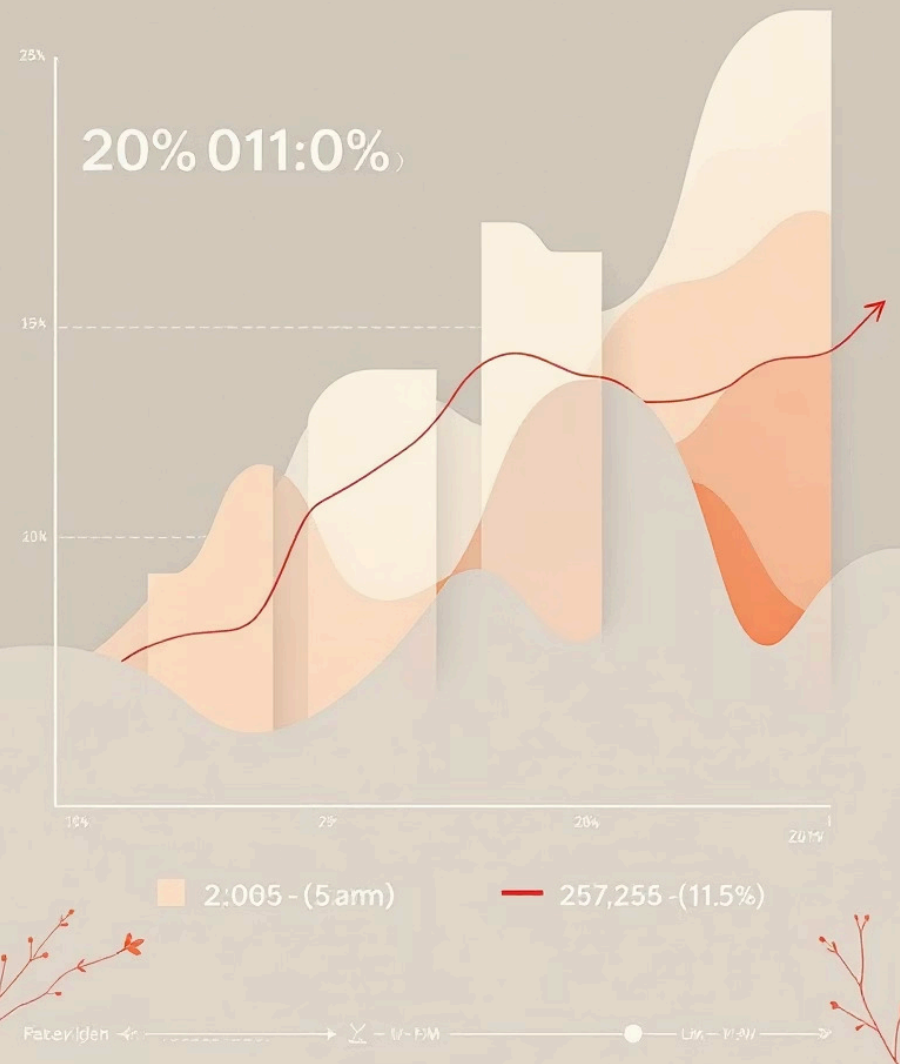
- **Tree Structure**

Decision Tree interpretability diagram





Performance Chart



Model Performance Comparison

Random Forest

Robust performance across all metrics

Decision Tree

High interpretability and transparency

Logistic Regression

Stable results with linear approach



Conclusion

Machine learning models effectively classify breast cancer tumors with high accuracy



Random Forest leads in robust performance



Decision Trees offer clear interpretability



Logistic Regression delivers consistent stability